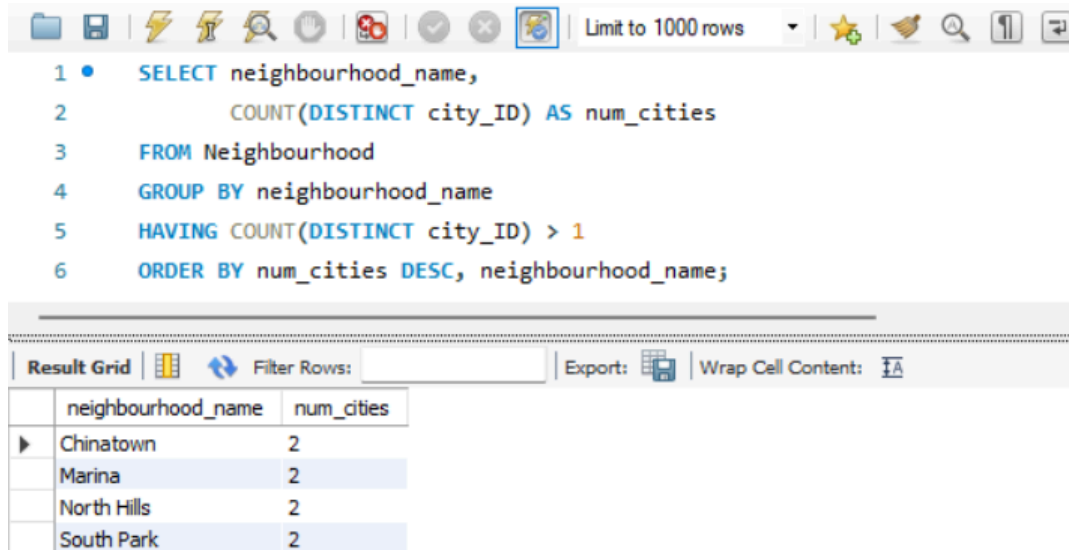


Question #1: Which neighbourhood names appear in more than one city?

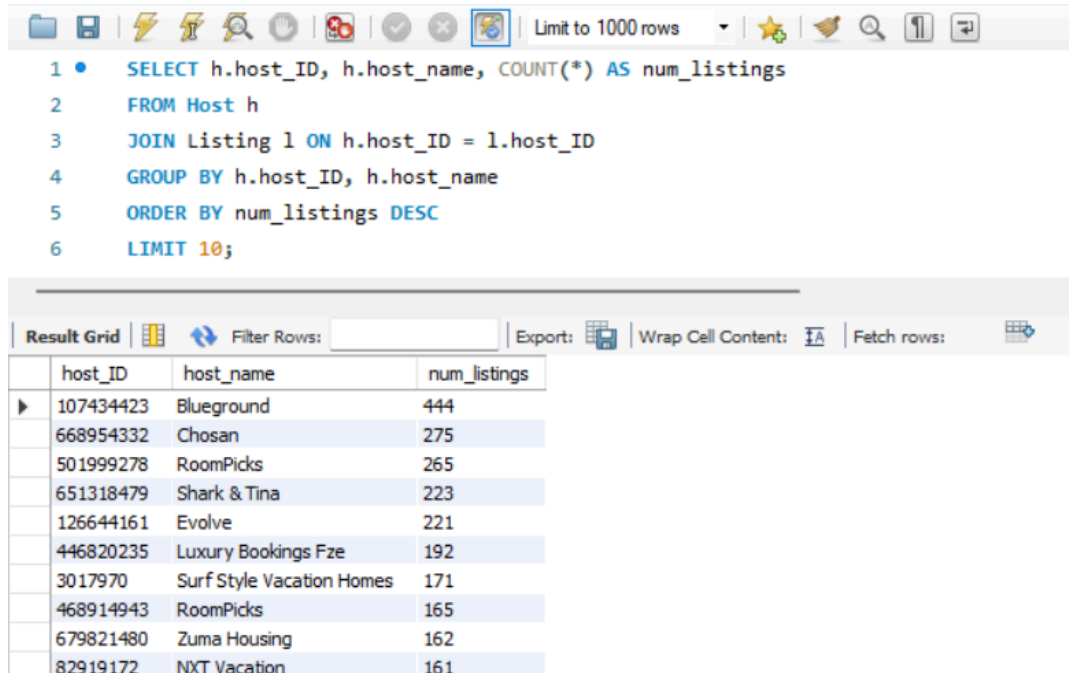


```
1 • SELECT neighbourhood_name,
2     COUNT(DISTINCT city_ID) AS num_cities
3 FROM Neighbourhood
4 GROUP BY neighbourhood_name
5 HAVING COUNT(DISTINCT city_ID) > 1
6 ORDER BY num_cities DESC, neighbourhood_name;
```

neighbourhood_name	num_cities
Chinatown	2
Marina	2
North Hills	2
South Park	2

This query's results justify our creation of the "neighbourhood_id" key that we use to identify different neighborhoods in cities, as it shows that multiple neighborhood names appear in more than one city. If we only identified different neighborhoods by their name, then listings from different cities would have been merged into one group.

Question #2: Across the three cities, who are the top 10 hosts with the most listings?

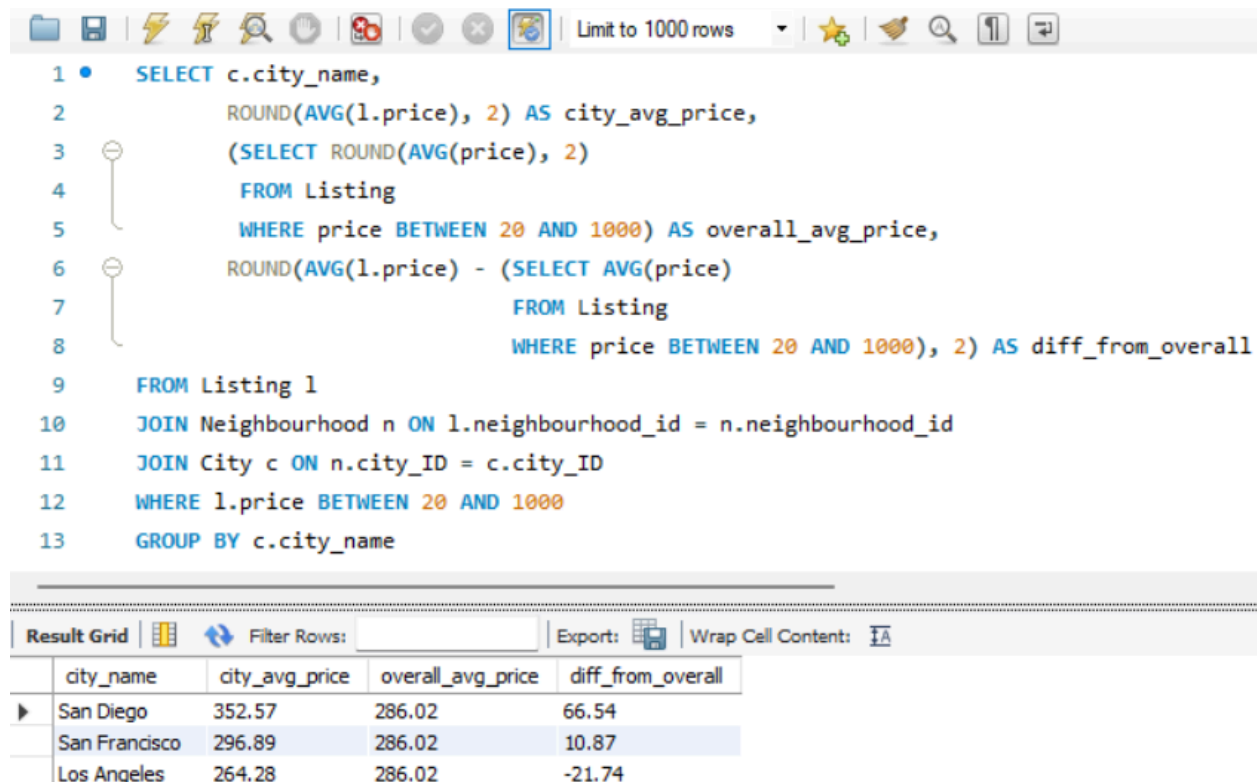


```
1 • SELECT h.host_ID, h.host_name, COUNT(*) AS num_listings
2 FROM Host h
3 JOIN Listing l ON h.host_ID = l.host_ID
4 GROUP BY h.host_ID, h.host_name
5 ORDER BY num_listings DESC
6 LIMIT 10;
```

host_ID	host_name	num_listings
107434423	Blueground	444
668954332	Chosan	275
501999278	RoomPicks	265
651318479	Shark & Tina	223
126644161	Evolve	221
446820235	Luxury Bookings Fze	192
3017970	Surf Style Vacation Homes	171
468914943	RoomPicks	165
679821480	Zuma Housing	162
82919172	NXT Vacation	161

From this query, we learn that the top 10 hosts can have hundreds of listings. These hosts are likely to be businesses, which we can discern from names like "Surf Style Vacation Homes" and "Zuma Housing." This query also reveals the importance of host_ID as the primary key of the Host table, as hosts can have identical names (e.g. "RoomPicks").

Question #3: How does each city's average price compare to the overall average?



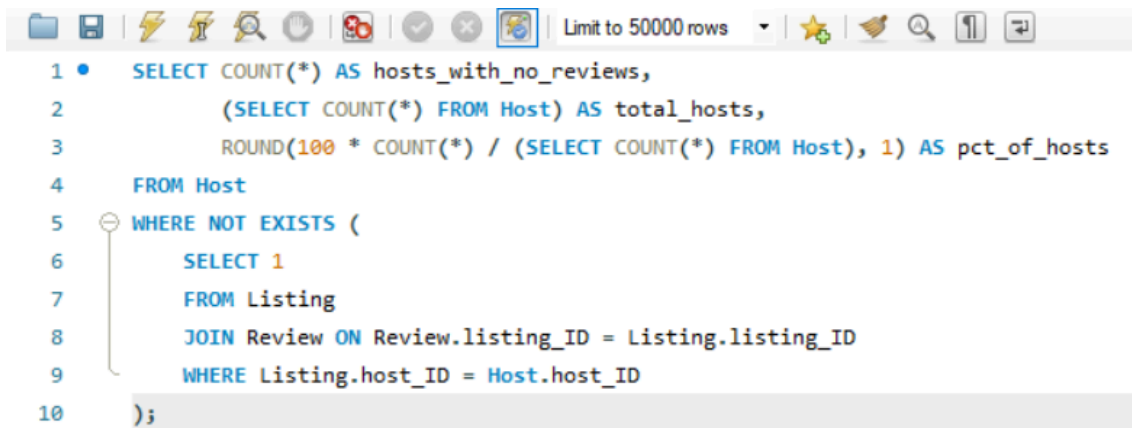
```
1 • SELECT c.city_name,
2       ROUND(AVG(l.price), 2) AS city_avg_price,
3       (SELECT ROUND(AVG(price), 2)
4       FROM Listing
5       WHERE price BETWEEN 20 AND 1000) AS overall_avg_price,
6       ROUND(AVG(l.price) - (SELECT AVG(price)
7       FROM Listing
8       WHERE price BETWEEN 20 AND 1000), 2) AS diff_from_overall
9 FROM Listing l
10 JOIN Neighbourhood n ON l.neighbourhood_id = n.neighbourhood_id
11 JOIN City c ON n.city_ID = c.city_ID
12 WHERE l.price BETWEEN 20 AND 1000
13 GROUP BY c.city_name
```

city_name	city_avg_price	overall_avg_price	diff_from_overall
San Diego	352.57	286.02	66.54
San Francisco	296.89	286.02	10.87
Los Angeles	264.28	286.02	-21.74

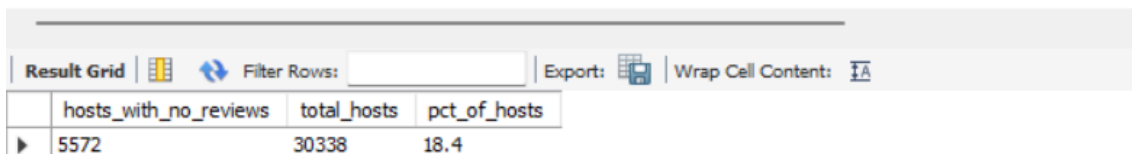
This query compares each city's average nightly price to the overall average across all three cities. Prices are limited to \$20-1,000 per night to prevent any extreme values skewing the average. The overall average is \$286.02. SD is the most expensive city at \$352.57, about 23% above the overall average. SF is slightly above at \$296.89 (about 4%), and LA is below at \$264.28 (about 8%).

LA contributes about 2/3 of all listings, so it pulls the baseline towards its own average price. It would be more fair to compare the other two cities against the rest of the data without LA. This result also doesn't explain why SD is higher. Differences could come from the mix of room types, since entire homes cost more than just rooms. The prices are also a snapshot from the scrape date and nightly prices could change with the seasons, so we cannot be entirely certain that this is the typical trend of average listing prices across the cities.

Question #4: How many hosts have never had a review on any of their listings?



```
1 • SELECT COUNT(*) AS hosts_with_no_reviews,  
2     (SELECT COUNT(*) FROM Host) AS total_hosts,  
3     ROUND(100 * COUNT(*) / (SELECT COUNT(*) FROM Host), 1) AS pct_of_hosts  
4 FROM Host  
5 WHERE NOT EXISTS (  
6     SELECT 1  
7     FROM Listing  
8     JOIN Review ON Review.listing_ID = Listing.listing_ID  
9     WHERE Listing.host_ID = Host.host_ID  
10 );
```



	hosts_with_no_reviews	total_hosts	pct_of_hosts
▶	5572	30338	18.4

This query finds hosts who have never received a review on any of their listings. Of the 30,338 hosts in the database, 5,572 (18.4%) have had no reviews. This means about four in five hosts (81.6%) have at least one reviewed listing.

We can't tell from this query exactly why these hosts have no reviews, but some possible explanations include: new hosts who haven't had any guests yet at the time this data was scraped, inactive or removed listings, hosts with a single listing that never booked, or guests who stayed but didn't leave a review. Review counts are also only a rough proxy for bookings, since not every guest writes one.